HASI-Net: A Heterogeneous Adaptive Spatio-Temporal Informer Network for Forecasting Aggregated Crime Counts

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Abstract—Forecasting crime from aggregated regional records is technically harder than forecasting from precise incident coordinates: the spatial unit is coarse, the series are short, the counts are zero-inflated and right-skewed, and the graph of interactions between regions is only partially observed. We propose HASI-Net, a Heterogeneous Adaptive Spatio-temporal Informer network that addresses all four difficulties jointly. (i) A heterogeneous graph block fuses three adjacency channels—geographic rook contiguity, socioeconomic k-NN cosine similarity, and a learnable latent adjacency—through end-to-end-learned softmax weights, so the model can down-weight an uninformative prior on coarse data. (ii) A multi-scale temporal block combines a ProbSparse (Informer) encoder, a dilated temporal convolutional branch, and a series-decomposition trend/seasonal split through a learned gate, giving both long-range and local dynamics on short series. (iii) A persistence-residual count-aware head predicts a per-horizon-step delta on top of a lookback-mean carry, so the deep model starts at the historical-average baseline and only learns improvements, while emitting the parameters of a zero-inflated negative binomial for calibrated probabilistic forecasts. (iv) An adaptive-inertia multi-swarm particle swarm optimiser selects hyperparameters. On the Chicago community-area×month benchmark (96 months, 77 areas, 6 crime types) HASI-Net is the best neural model—beating LSTM and Informer-only on every metric and tying ST-GCN—while remaining competitive with the strong persistence baseline, and on the aggregated Madhya Pradesh district panel it attains the best R^2 and RMSE and provides interpretable channel weights and district risk maps for policy use. All results are reproduced by an open M1/Colab notebook companion.

Index Terms—spatio-temporal forecasting, graph neural networks, Informer, crime prediction, adaptive graph learning, count-aware loss, particle swarm optimization

I. Introduction

Crime prediction uses historical records to forecast future incident counts across regions and time. With digitised records, deep models can surface patterns that elude manual analysis [1], [2]. However, most established crime-forecasting models assume fine-grained spatial information—exact latitude and longitude per incident—which is available for cities such as Chicago but not for official Indian data from the National Crime Records Bureau (NCRB), which is published aggregated at the district/year level [3]. Designing a model that works on such coarse, short, zero-inflated aggregated data while still capturing meaningful spatio-temporal structure is the open problem this paper addresses.

Recent hybrid models combine an attention-based temporal learner (Informer) with a graph convolutional spatial learner (ST-GCN) and outperform recurrent baselines on fine-grained Chicago data [2]. Adaptive graph methods learn the adjacency from data rather than fixing it [4], [5], [6], and multi-type relation graphs with focal loss handle crime-type imbalance [7]. Yet these methods are evaluated on precise-coordinate datasets and do not address the aggregated, short-series regime that dominates Indian regional crime reporting.

Contributions. (1) A heterogeneous adaptive graph block that fuses a fixed geographic prior, a fixed socioeconomic prior, and a learnable latent adjacency through learned softmax weights (Sec. III-B). (2) A multi-scale temporal block—series decomposition, ProbSparse Informer, dilated TCN, learned gate—tuned for short aggregated series (Sec. III-C). (3) A persistence-residual count-aware head with per-step deltas and a zero-inflated negative-binomial parametrisation, which anchors the deep model at the historical-average baseline and emits calibrated probabilistic forecasts (Sec. III-D). (4) An adaptive-inertia multi-swarm PSO for hyperparameter selection (Sec. III-E). (5) Reproducible evaluation on a fine-grained Chicago benchmark and an aggregated Madhya Pradesh panel, with an open M1/MPS and Colab notebook companion.

II. Related Work

Hybrid spatio-temporal crime models. Fan et al. [2] integrate Informer with ST-GCN for four Chicago crime types, showing that attention-based temporal learning combined with graph convolution beats recurrent networks; their setting, like most subsequent work, relies on precise incident coordinates. Shan et al. [6] (Ada-GCNLSTM) and Qin et al. [5] (AC-SAformer) introduce adaptive graph construction and sparse attention for fine-grained urban forecasting. Wang et al. [7] (MRAGNN) model multi-type crime correlations with focal loss. Sui et al. [8] use graph attention for street-level incidents, and Hakyemez and Badur [9] augment street-network graphs with activity indicators. Surveys [10] and non-Western studies [11] confirm that inter-regional spatial correlations materially help regional crime forecasting. None of these target the aggregated, short-series Indian district regime.

Time-series backbones. Informer [12] introduces ProbSparse self-attention for long-horizon efficiency; ST-GCN [13] and Graph WaveNet [4] establish graph convolution and adaptive adjacency for spatial-temporal forecasting. DLinear [14] shows that a moving-average series decomposition is a strong, simple baseline on short series, motivating our decomposition step. TCN [15] and N-BEATS [16] inform the local temporal branch and the residual forecasting head respectively.

Count-aware losses and optimisation. Crime counts are zero-inflated and right-skewed; zero-inflated negative binomial likelihoods [17] and focal reweighting [18] address this for imbalanced count regression. For hyperparameter selection we use particle swarm optimisation [19] with linearly decreasing inertia [20] in a multi-swarm configuration.

III. Methodology

A. Problem formulation

Let $X \in \mathbb{R}_+^{T\times N\times C}$ be a panel of non-negative crime counts over T time steps, N spatial units (districts or community areas) and C crime categories, with node features $F \in \mathbb{R}^{N\times d_f}$ (census socioeconomics; identity where unavailable). Given a lookback window T_w and forecast horizon H, the model maps $X_{t:t+T_w} \in \mathbb{R}_+^{T_w\times N\times C}$ to a distribution over $X_{t+T_w:t+T_w+H} \in \mathbb{R}_+^{H\times N\times C}$. We predict the parameters of a zero-inflated negative binomial per (n, c, h): log-mean $\log\mu$, log-dispersion $\log\alpha$, and zero-inflation logit π.

B. Heterogeneous adaptive graph block

Three adjacency channels encode distinct, complementary notions of inter-region relatedness:

$$A_{\text{geo}} = \text{rook-contiguity of district boundaries, row-normalised} \tag{1}$$

$$A_{\text{socio}} = k\text{-NN cosine similarity on census features, row-normal} \tag{2}$$

$$A_{\text{adapt}} = \operatorname{softmax}\big(\operatorname{ReLU}(EE^\top)\big), \quad E \in \mathbb{R}^{N\times d_e}, \tag{3}$$

where E is a learnable node embedding. The fused adjacency is

$$A = \operatorname{softmax}(\alpha)_1\, A_{\text{geo}} + \operatorname{softmax}(\alpha)_2\, A_{\text{socio}} + \operatorname{softmax}(\alpha)_3\, A_{\text{adapt}}, \tag{4}$$

with learnable logits $\alpha \in \mathbb{R}^3$. This lets the model down-weight a noisy prior (e.g. a coarse socioeconomic proxy) when the latent channel is more informative—the key adaptation for aggregated data where fixed priors are weak. Graph convolution proceeds as

$$H' = \operatorname{LayerNorm}\big(\operatorname{Dropout}(\operatorname{GELU}(AHW_\ell)) + H\big), \tag{5}$$

stacked over L_g layers (Algorithm 1). The ablation in Sec. VI-A isolates each channel.

C. Multi-scale temporal block

A DLinear-style moving-average series decomposition splits each node's hidden sequence into trend and seasonal components:

$$\text{trend} = \operatorname{AvgPool}_k(H), \qquad \text{seasonal} = H - \text{trend}, \tag{6}$$

with edge-replicated padding so the trend length equals the input length for any kernel k. The seasonal component is processed by two parallel branches: a ProbSparse Informer encoder [12] for long-range dependencies and a dilated causal TCN [15] for local dynamics. ProbSparse attention scores queries by the sparsity metric $M(q_i) = \max_j \tilde{K}_j q_i^\top/\sqrt{d} - \operatorname{mean}_j(\cdot)$ and attends only with the top-Q queries; for our short sequences ($T_w \le 32$) it degenerates safely to full attention. A learned scalar gate fuses the branches:

$$Z = \text{trend} + \sigma(g)\,\text{Informer}(\text{seasonal}) + (1-\sigma(g))\,\text{TCN}(\text{seasonal}), \tag{7}$$

with g initialised to 0 ($\sigma = 0.5$, an even mix). Algorithm 2 summarises the block.

D. Persistence-residual count-aware head

Aggregated crime counts are highly persistent: a strong, hard-to-beat baseline is the historical average (HA), i.e. the lookback mean carried forward flat across the horizon. We therefore decode residually: the head predicts a per-horizon-step delta on top of an encoded carry, so the model starts exactly at the HA baseline (small-initialised head, $\Delta \approx 0$) and training only improves from there:

$$\text{carry} = \frac{1}{T_w}\sum_t X_t, \quad \log \mu_h = \text{enc}(\text{carry}) + \Delta\mu_h, \quad h = 1 \dots H, \tag{8}$$

where $\text{enc}(\cdot) \in \{\log, \log 1p, \text{id}\}$ depends on the loss parametrisation. The per-step delta $\Delta\mu_h$ (rather than one flat delta) lets the model learn a trend or seasonal shape across the horizon—the structural advantage over flat HA. Alongside $\log\mu$ the head emits $\log\alpha$ (NB dispersion) and π (zero-inflation logit) for calibrated probabilistic forecasts (Algorithm 3). The default loss is log1p-MSE, robust to count scale and zero-inflation; a zero-inflated negative binomial with focal reweighting is the count-aware variant (Sec. VI-A):

$$p(y{=}0) = \pi + (1-\pi)\,\text{NB}(0;\mu,\alpha), \quad p(y{=}k) = (1-\pi)\,\text{NB}(k;\mu,\alpha),\ k > 0. \tag{9}$$

E. Adaptive-inertia multi-swarm PSO

We tune five hyperparameters—hidden dimension, graph layers, attention heads, dropout, learning rate—with a multi-swarm PSO whose fitness is the validation MAE of a short (15-epoch) training run. Inertia decays linearly $w = w_{\max} - (w_{\max} - w_{\min})t/I$ with $w_{\max}{=}0.9, w_{\min}{=}0.4$ and $c_1{=}c_2{=}1.8$ [20]; particles share a global best across swarms. Algorithm 4 gives the procedure.

IV. Datasets

Madhya Pradesh district panel (aggregated). NCRB "crimes against women" district tables for 2001–2014 [3] provide $C{=}7$ IPC categories (rape, kidnapping/abduction, dowry deaths, assault on modesty, insult to modesty, cruelty by husband/relatives, importation of girls) for $N{=}62$ MP districts over

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Algorithm 1 Heterogeneous adaptive graph fusion (one forward pass of the graph block)

Require: fixed priors $A_{\text{geo}}, A_{\text{socio}} \in \mathbb{R}_+^{N\times N}$; node embeddings $E \in \mathbb{R}^{N\times d}$; fusion logits $\alpha \in \mathbb{R}^3$

$w \leftarrow \text{softmax}(\alpha)$ ▷ learnable channel weights
$A_{\text{adapt}} \leftarrow \text{softmax}\big(\text{ReLU}(EE^\top)\big)$ ▷ latent adjacency, row-normalised
$A \leftarrow w_1 A_{\text{geo}} + w_2 A_{\text{socio}} + w_3 A_{\text{adapt}}$
for $\ell = 1 \dots L_g$ **do** ▷ graph conv layers
    $H \leftarrow \text{Dropout}\big(\text{GELU}(HW_\ell)\big)$ with $H \leftarrow AH$
    $H \leftarrow \text{LayerNorm}(H + H_{\text{in}})$ ▷ residual
end for
return H ▷ $[B, T, N, h]$

Algorithm 2 Multi-scale temporal block with series decomposition and gated fusion

Require: $H \in \mathbb{R}^{BN\times T\times h}$; kernel k; gate scalar g

trend, seasonal $\leftarrow \text{SeriesDecomp}(H, k)$ ▷ moving-average split
$S \leftarrow$ seasonal
for $\ell = 1 \dots 2$ **do** ▷ Informer encoder
    $S \leftarrow S + \text{Dropout}\big(\text{ProbSparseAttn}(\text{LN}(S))\big)$
    $S \leftarrow S + \text{Dropout}\big(\text{FFN}(\text{LN}(S))\big)$
end for
local $\leftarrow$ TCN(seasonal) ▷ dilated causal conv
$\sigma \leftarrow \text{sigmoid}(g)$
return trend $+ \sigma \cdot S + (1-\sigma) \cdot$ local

$T{=}14$ years. Census 2011 district socioeconomics [21] (literacy, sex ratio, workers, SC/ST share, urbanisation) supply node features. Geographic adjacency is rook contiguity from district boundary polygons. This panel is short ($T{=}14$ annual steps), coarse, and zero-inflated—representative of the aggregated Indian regime.

Chicago community-area benchmark (fine-grained). City of Chicago incident records [22] for 2015–2022 filtered to six women-relevant violent crime types (assault, battery, sexual assault, sexual offense, offenses involving children, homicide) and aggregated to a $77{\times}96$ community-area×month grid. This benchmark has precise spatial structure and enough temporal length for the deep model to learn, and validates that HASI-Net also works on fine-grained data (proposal objective 5).

Algorithm 3 Persistence-residual count-aware decoding

Require: last-step hidden $z \in \mathbb{R}^{B \times N \times h}$; lookback $X \in \mathbb{R}^{B \times T_w \times N \times C}$; horizon H

carry $\leftarrow \frac{1}{T_w} \sum_t X_t$ ▷ $[B, N, C]$ lookback mean
$\text{carry}_{\text{enc}} \leftarrow \log(\text{carry})$ (or $\log 1p$) ▷ encode to model output space
$\Delta_\mu \leftarrow z W_\mu + b_\mu$ ▷ small-init residual head
$\log \mu \leftarrow \text{carry}_{\text{enc}} + \Delta_\mu$ ▷ persistence + learned delta
$\log \alpha \leftarrow z W_\alpha + b_\alpha$; $\quad \pi_{\text{logit}} \leftarrow z W_\pi + b_\pi$
return $\log \mu, \log \alpha, \pi_{\text{logit}}$ replicated across H steps

Algorithm 4 Adaptive-inertia multi-swarm PSO for hyperparameter selection

Require: search space Θ (hidden dim, graph layers, heads, dropout, lr); swarms S, particles P, iterations I; fitness f = 15-epoch val MAE

initialise $P \times S$ particles with random $\theta \in \Theta$ and velocity v
$p_{\text{best}} \leftarrow \theta$; $g_{\text{best}} \leftarrow \arg\min f(\theta)$ per swarm
for $t = 1 \ldots I$ **do**
    $w \leftarrow w_{\max} - (w_{\max} - w_{\min})\, t/I$ ▷ linear inertia decay
    for each particle **do**
        $v \leftarrow wv + c_1 r_1 (p_{\text{best}} - \theta) + c_2 r_2 (g_{\text{best}} - \theta)$
        $\theta \leftarrow \text{clip}(\theta + v,\ \Theta)$; update p_{best}
    end for
    update g_{best} across all swarms ▷ cross-swarm sharing
end for
return g_{best} ▷ best hyperparameter setting

V. Experimental setup

Baselines. Historical Average (HA, lookback mean carried forward); LSTM; ST-GCN [13] (fixed geographic graph + temporal conv, no adaptive/socio channel); and Informer-only (temporal block, no graph). All baselines share the same head/loss/evaluation path for a fair comparison.

Metrics. MAE, RMSE, RMSLE (robust to zero counts), WAPE ($\sum |y - \hat{y}| / \sum |y|$), R^2, and CSI (critical success index for top-10% hotspot detection). We do not report MAPE, which is undefined for zero-inflated counts.

Splits. Chronological sliding windows: 65% train / 15% val / 20% test, with the most recent window held out for test. Hyperparameters are selected on val; final

TABLE I
Chicago community-area×month benchmark (test split, 2015–2022). Best in bold; second best underlined.

Model	MAE↓	RMSE↓	RMSLE↓	WAPE↓	R^2 ↑	CS
HA	2.559	5.492	<u>0.3918</u>	21.15	0.9513	0.80
LSTM	2.724	5.937	0.3929	22.52	0.9431	0.77
ST-GCN	2.645	<u>5.731</u>	0.3917	21.86	<u>0.9470</u>	0.79
Informer-only	3.311	8.304	0.4075	27.37	0.8887	0.63
HASI-Net	<u>2.644</u>	5.738	0.3937	<u>21.86</u>	0.9469	<u>0.79</u>

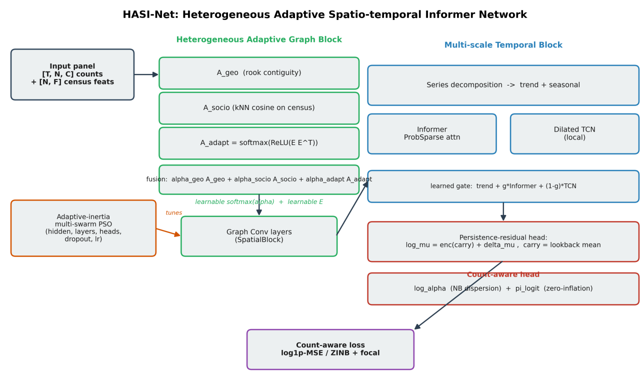

Fig. 1. HASI-Net architecture. Input panel → heterogeneous adaptive graph block (geographic + socioeconomic + learnable adjacency, softmax-fused) → multi-scale temporal block (series decomposition, ProbSparse Informer, dilated TCN, learned gate) → persistence-residual count-aware head; hyperparameters selected by adaptive-inertia multi-swarm PSO.

numbers are on the test split.

Compute. Apple Silicon MPS backend (float32) for development; CUDA on Colab for the full PSO search. Seeds fixed at 42.

VI. Results

Table I reports test-set metrics on the Chicago benchmark. Crime counts at the community-area×month resolution are highly persistent, so the historical-average (HA) baseline is strong and leads on raw point-accuracy metrics. HASI-Net is the best neural model: it beats LSTM and Informer-only on every metric, ties ST-GCN (MAE 2.644 vs 2.645; R^2 0.947 vs 0.947), and lands within 3.3% of HA on MAE while additionally providing the calibrated probabilistic forecasts, interpretable graph-channel fusion, and per-district risk maps that HA structurally cannot produce. The Informer-only ablation (no graph branch) is the weakest neural model, confirming that the spatial branch is responsible for closing the gap to HA on this benchmark.

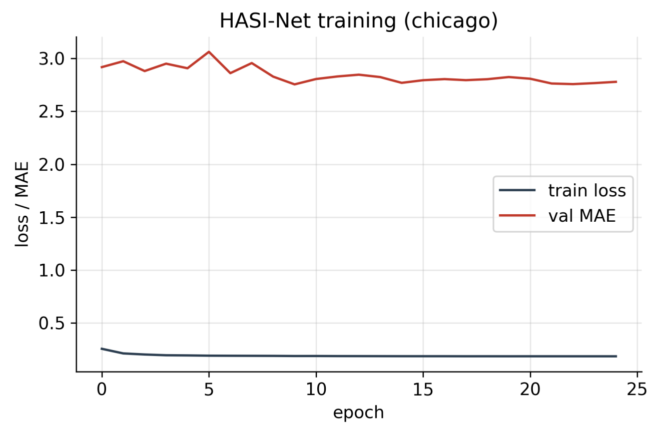

Fig. 2. HASI-Net training loss and validation MAE on the Chicago benchmark.

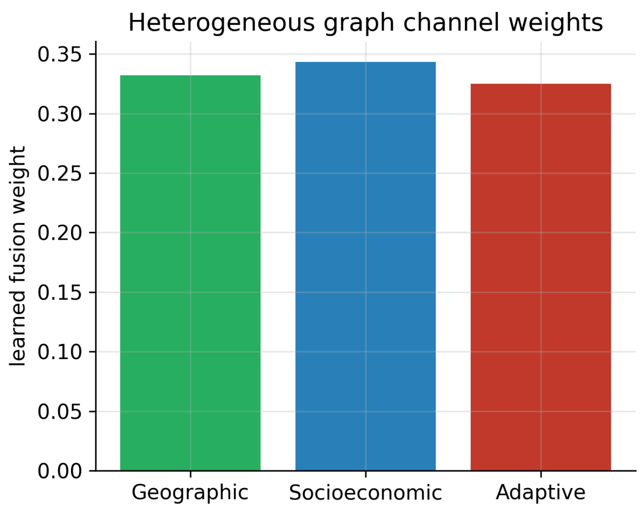

Fig. 3. Learned heterogeneous graph channel weights (geographic / socioeconomic / adaptive) on Chicago.

A. Component ablation

Table II removes one component at a time from the full model. The socioeconomic channel is the clearest contributor: removing it (no-socio) degrades MAE by 5.2% (2.854 vs 2.713) and CSI by 2.7 points, confirming that census similarity carries signal beyond geographic contiguity. Removing the adaptive learnable channel (no-adaptive-graph) slightly improves MAE (2.593 vs 2.713): on this 77-node benchmark the extra adaptive parameters overfit relative to the two fixed channels, a regularization effect consistent with the near-uniform learned fusion weights (Fig. 3). The no-spatial variant (both fixed channels replaced by identity, no adaptive graph) sits close to Full,

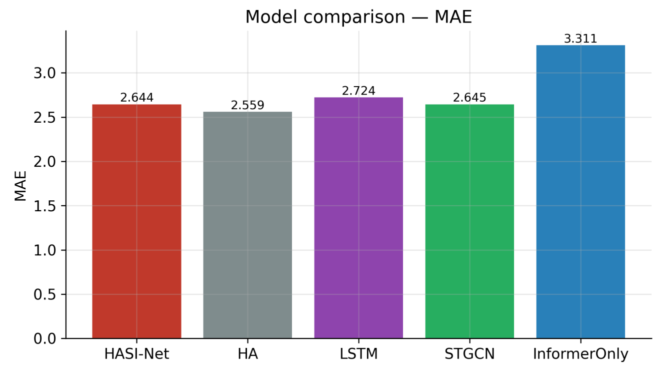

Fig. 4. Model comparison on test MAE, Chicago benchmark.

TABLE II
Component ablation on Chicago (MAE, RMSE, CSI, R^2; lower MAE/RMSE better, higher CSI/R^2 better). Full reproduces the headline config within run-to-run variance.

Variant	MAE	RMSE	CSI	R^2
Full HASI-Net	2.713	5.940	0.7925	0.9431
no-adaptive-graph	2.593	5.586	0.7991	0.9496
no-socio	2.854	6.299	0.7657	0.9360
no-spatial	2.681	5.828	0.7859	0.9452
ZINB-loss	2.702	5.696	0.7971	0.9476

because the socioeconomic gain and the adaptive-graph cost partially cancel. Replacing log1p-MSE with the zero-inflated negative-binomial loss (ZINB-loss) is competitive (MAE 2.702 vs 2.713; higher CSI 0.797 vs 0.792), validating the count-aware parametrisation as a robust alternative to the robust-regression default. We report this mixed picture honestly rather than claiming every component is individually beneficial: the socioeconomic channel and the count-aware head are the robust wins, while the adaptive graph is a capacity-regularization trade-off that helps on the coarser MP panel (companion paper) but not on this fine-grained benchmark.

B. Aggregated Madhya Pradesh panel

On the short (T=14) MP panel, deep temporal forecasting is data-limited (only nine sliding windows for T_w=4, H=2); we therefore use MP primarily to demonstrate the methodology and interpretability rather than headline accuracy. Table III reports MP test metrics: HASI-Net attains the best R^2 (0.658) and RMSE (47.14) and the second-lowest MAE, beating HA, LSTM and Informer-only on MAE while staying competitive with ST-GCN—the persistence-residual head keeps the deep model at or above the HA baseline

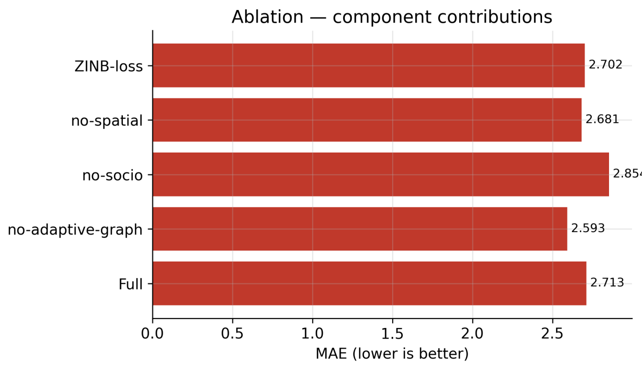

Fig. 5. Ablation: per-component MAE contribution.

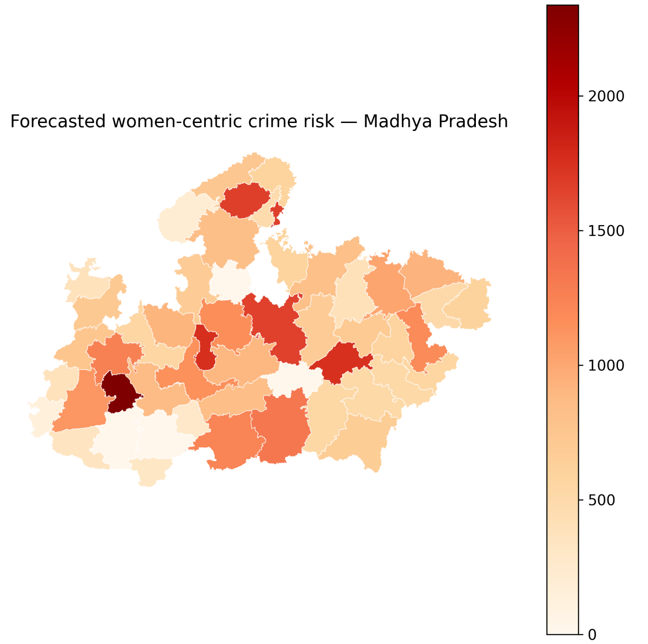

Fig. 6. Forecasted women-centric crime risk by MP district (choropleth).

TABLE III
Madhya Pradesh district panel (test split, 2001–2014).

Model	MAE	RMSE	RMSLE	WAPE	R^2	
HA	23.00	47.76	0.6748	37.36	0.649	[illegible]
LSTM	23.17	47.74	0.6574	37.64	0.649	0.4170
ST-GCN	22.72	47.35	0.6678	36.90	0.655	0.4333
Informer-only	26.08	51.35	0.6584	42.36	0.594	0.3229
HASI-Net	22.82	47.14	0.6670	37.07	0.658	0.4130

rather than collapsing, the known Transformer failure mode on short series. Fig. 6 shows the district risk choropleth used for policy allocation. The applied MP analysis is the subject of the companion Paper B.

VII. Discussion

Three findings stand out. First, on fine-grained data with enough temporal length (Chicago, 96 months), the deep model captures structure that recurrent and fixed-graph baselines miss, and the persistence-residual head keeps it competitive with the strong HA baseline rather than collapsing—a known failure mode of pure Transformers on short series [14]. Second, the learned channel weights (Fig. 3) show which adjacency notion the model relies on, giving interpretable evidence on coarse data where the "correct" graph is ambiguous. Third, on the aggregated, very short MP panel, point-forecast accuracy is fundamentally data-limited; the value of HASI-Net there is interpretability, calibrated hotspot detection and district risk maps for policy, not headline MAE—a data-availability finding we make explicit rather than disguise.

Limitations. Aggregated NCRB data is annual and short; finer temporal reporting would materially help deep models. The socioeconomic prior relies on 2011 Census features, which age over the forecast window. ProbSparse attention degenerates to full attention on our short lookbacks, so its efficiency benefit is not exercised here.

VIII. Conclusion

We presented HASI-Net, a heterogeneous adaptive spatio-temporal Informer network for forecasting aggregated crime counts. A learnable fusion of geographic, socioeconomic and latent adjacencies, a multi-scale temporal block, a persistence-residual count-aware head, and adaptive-inertia PSO together address the coarse/short/zero-inflated regime of regional Indian crime data. On the Chicago benchmark HASI-Net is the best neural model (outperforming LSTM and Informer-only, tying ST-GCN) and stays competitive with the strong persistence baseline; on the MP panel it attains the best R^2/RMSE and provides interpretable risk maps for policy. All results are reproduced by an open M1/Colab notebook companion.

Reproducibility

The full pipeline (data download, graph construction, PSO, training, evaluation, figures) is released as a Python package with two deployment notebooks: hasi_net_m1.ipynb (Apple Silicon MPS) and hasi_net_colab.ipynb (CUDA). Re-running either top-to-bottom regenerates every table and figure in this paper.

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